

How the literature models the benefit and the cost of children, and what it implies for P1

Fertility–housing project, decision note

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Purpose

The consolidated review (P1) found that the model’s benefit of children is linear in children at home while its cost is concave, so that a household without taste shocks wants zero children or the biological maximum, and the one-child and three-plus shares are carried by the continuation taste scale. This note does the reading that choice deserves. Every functional form below was read from the paper’s own PDF by a verifier and quoted with a page; papers not on disk are marked and cited only for a headline claim. The July 18 readings memo (equivalence_scales_sequential_fertility_readings_20260718.tex) already covers the equivalence-scale literature (Scholz–Seshadri–Khitatrakun, Fernández-Villaverde–Krueger, Attanasio et al., Comin–Lashkari–Mestieri) and is not repeated.

Notation: m children at home, n children ever born, $X = c^\alpha(s - \bar{h})^{1-\alpha}$ the composite, $e(m) = ((2 + 0.7m)/2)^{0.7}$ the scale, $\sigma = 2$. The model’s flow utility is $u = -e(m)/X + \psi m$.

1. What the model does, in one line each

Benefit: ψm , a constant flow ψ per child per period while the child is at home. Cost: the scale $e(m)$ deflating the composite, so at $\sigma = 2$ the cost of the k -th child in units of $1/X$ is $e(k) - e(k-1) = 0.234, 0.216, 0.203$, falling in k ; plus a parenthood-only space floor $\bar{h}(m) = h_P \mathbf{1}\{m > 0\}$, which costs only at the first child; plus a one-time utility cost ξ at the first birth. Hence the net flow benefit of the k -th child rises in k . No income gradient is built in: the scale is homothetic and the floor is an absolute quantity, so relative to income the cost of children falls as income rises.

2. The benefit side: three families

Concave in the number ever born. Barro and Becker (1989, p. 2): $U_i = v(c_i) + a(n_i) n_i U_{i+1}$ with $a(n) = a n^{-\varepsilon}$, so the weight on children is $n^{1-\varepsilon}$, concave. Sommer (2016, p. 34): $U = c^{1-\gamma}/(1 - \gamma) + \zeta(nq)^{1-\kappa}/(1 - \kappa)$, concave in n at given quality, entering every period the children are at home. De la Croix and Doepke (2003, p. 2): $\ln c + \beta \ln d' + \gamma \ln(nh')$. Baudin, de la Croix and Gobbi (2015, p. 1861): $\ln c + \ln(n + \nu)$, with $\nu > 0$ so that childlessness is defined. Kim, Tertilt and Yum (2024, p. 1587): $\phi(n) \log(h' - \chi \tilde{h}')$ with $\phi(n)$ nonparametric for $n = 1, 2, 3$. Daruich and Kozłowski (2020, p. 226): $b(n)u(c_k)$, b increasing and concave. The handbook exposition (Doepke, Hannusch, Kindermann

and Tertilt 2023, p. 161) uses $\log c + \delta \log(nh)$. This is the majority form: diminishing marginal benefit gives an interior family size from preferences alone.

Linear, one-time at birth. Doepke and Kindermann (2019, p. 3284): $u = c - d + vb$, utility from a child only in the period of birth, with v heterogeneous across couples and a veto rule. Here the intensive margin comes from heterogeneity in v and from per-child costs, not from curvature. Moreno-Maldonado and Santamaría (2024, p. 17) likewise put an idiosyncratic multiplicative child taste η^i on utility and choose among young, postponed and never.

Linear flow while at home. This is the model's form. I found no paper on disk that uses exactly it. The closest are the two linear one-time forms above, and both pair linearity with per-child costs that are linear or convex in the number of children.

The lesson is not that linear is wrong; it is that linear benefits are always paired in the literature with either heterogeneity in the taste (Doepke–Kindermann, Moreno-Maldonado–Santamaría) or costs that do not fall in k . The model has the taste heterogeneity (the fertility shocks) but pairs it with falling costs.

3. The cost side: four devices and their curvature in k

Time cost proportional to the wage. De la Croix and Doepke (2003, eq. 1): income $wh(1 - \phi n)$, $\phi = 0.075$ per child. Baudin et al. (2015): a fixed time cost η of becoming a parent plus ϕ per child, no goods cost. Kim, Tertilt and Yum (2024, eq. 7): λn of parental time plus education spending xn . Doepke and Kindermann (2019): the mother's foregone wage or the childcare price while a child is under three, plus a fixed goods cost ϕ_c and a fixed utility cost ϕ_u per child. Moreno-Maldonado and Santamaría embed an estimated child earnings penalty in the income process. Curvature: linear in n (constant marginal cost), and proportional to the wage, which is what produces the negative income–fertility gradient (Jones, Schoonbroodt and Tertilt 2010, not on disk, as cited by Baudin et al. fn. 16: with a log form and no non-labor income the gradient needs a wage-proportional cost).

Goods cost per child. Barro–Becker: P_i per child. Couillard (2025, eq. 4.1): $c + ni + nrh = y$, a per-child goods investment and a per-child housing cost rh per child. Curvature: linear in n .

Equivalence scale. Two versions. Deflating consumption only, $U(c/e)$: Kim, Tertilt and Yum (p. 1588, OECD-modified weights), Borella–De Nardi–Yang (divisor $(j + 0.7f)^{0.7}$), and this model. Multiplying utility as well, $eU(c/e)$: Scholz, Seshadri and Khitatrakun (2006, p. 615, $E \sum \beta^{j-S} n_j U(c_j/n_j)$). Curvature at $\sigma = 2$ with the Citro–Michael scale: deflating only gives cost increments 0.234, 0.216, 0.203, falling; multiplying gives $-e^2/X$, increments 0.522, 0.580, 0.630, rising. The sign of the curvature flips with the outer weighting. Neither SSK nor Borella et al. choose fertility, so neither paper had to confront this; Kim, Tertilt and Yum do choose fertility and pair the deflating scale with a concave $\phi(n)$ and linear time and education costs.

Minimum quality or space. Sommer (2016, p. 5): $q \geq \bar{q}$ if $n > 0$, with quality produced from time and goods, so the floor costs $n\bar{q}$ worth of inputs, linear in n . Couillard: rh per child, linear. This model: h_P once, at the first child, with the per-child slope restricted to zero in the current vintage. Curvature: the literature's floors are per child; ours is a step.

Summary of curvature. Every paper that chooses fertility has a marginal cost that is constant or rising in the number of children. Ours is the only one with a falling marginal cost, and it comes from the one device (a deflating scale under $\sigma > 1$ with exponent below one) that the fertility papers either pair with curvature elsewhere or do not use.

4. The income gradient

Cross-sectional fertility falls with income in the United States. In the models above this comes from a cost proportional to the wage (time cost or earnings penalty), or from quantity–quality with a high income elasticity of quality (Becker–Lewis, not on disk; the handbook’s eq. 2). A homothetic scale gives a flat gradient; an absolute floor gives a rising one. The model has neither device, and the E6b diagnostic on July 27 found the childlessness gradient across permanent types reversed relative to the CPS. This is the same symptom as P3’s ξ : a first-birth cost in utility units is income-neutral.

5. Three candidate specifications

Each is stated as primitives, with what it adds or removes and which moments identify it. None is implemented by this note.

S1. Concave benefit, everything else unchanged. Replace ψm by $\psi v(m)$, $v(m) = \log(1 + m)$ or $(1 + m)^{1-\varepsilon}$ with ε fixed externally. Net flow benefit of the k -th child then falls if v is concave enough to beat the falling scale cost; with $v = \log(1 + m)$ the benefit increments are 0.693, 0.405, 0.288, against cost increments 0.234, 0.216, 0.203, so the net benefit falls in k and an interior family size exists. Adds no parameter if ε is fixed. Identified by the one-child and three-plus shares, which then stop being κ_C ’s job. Does not fix the income gradient.

S2. The SSK weighting. Keep ψm , replace $U(X/e)$ by $e U(X/e)$, so flow utility is $-e(m)^2/X + \psi m$. Costs now rise in k ; an interior size exists with a linear benefit. Adds no parameter and matches the cited paper’s objective. Two costs. The first child’s utility cost more than doubles, so $\psi, \xi, \kappa_1, \kappa_C$ all move and the calibration restarts. And it puts the negative level of CRRA utility to work: multiplying a negative number by family size penalizes children through the level normalization, the same objection ChatGPT raised against child-scaled bequests; Daruich and Kozlowski (fn. 11) and Jones–Schoonbroodt–Tertilt note that Barro–Becker weighting needs positive utility to be meaningful. S2 is defensible only if the paper states this and shows the $\sigma = 2$ level is not driving the result. Does not fix the income gradient.

S3. A wage-proportional child cost. Keep the current benefit and scale; add a time or earnings cost while $m > 0$, $y^d(1 - \tau_c(m))$, with τ_c taken from the child-penalty literature (Kleven et al., not on disk; Moreno-Maldonado–Santamaría estimate it on U.S. data), and make the space floor per child as in Couillard. Marginal cost becomes constant in k and proportional to the wage, so the income gradient appears and the intensive margin is set by cost against a linear benefit plus shocks, the Doepke–Kindermann configuration. Adds one external schedule $\tau_c(m)$ and possibly retires ξ (P3). Identified by fertility and childlessness by income or education (CPS), which the model currently does not target, and by the three-plus versus one-to-two rooms gap for the per-child floor.

What I would do. S3 first, because it is the only one that answers both objections (curvature and gradient) with an externally measured object, and it is the configuration the two quantitative fertility references use. Then test whether S1’s curvature is still needed for the one-child share. S2 is a one-line change worth running as a diagnostic to show the author what the cited objective implies, not a baseline. Whatever is chosen, the paper’s preference paragraph must state the benefit form, the four cost devices in play, and why the marginal cost does not fall in the number of children.

6. What each reference actually assumes, for the paper’s citations

Paper	Benefit of children	Cost of children	Scale	Children as
Barro–Becker 1989	$n^{1-\varepsilon}U_{i+1}$, concave	goods P_i per child	none	ever born, once
Sommer 2016	$\zeta(nq)^{1-\kappa}$, concave, per period	time and goods into quality; floor $\bar{q}$ per child	none	at home, stock
Doepke–Kindermann 2019	$v b$, linear, at birth, v heterogeneous	fixed goods ϕ_c , fixed utility ϕ_u , mother’s wage under 3	none	birth flow; stock for care cost
de la Croix–Doepke 2003	$\gamma \ln(nh')$	time ϕn at own wage; education at teacher wage	none	ever born, once
Baudin et al. 2015	$\ln(n + \nu)$	time: fixed η for first child, ϕ per child	none	ever born
Kim–Tertilt–Yum 2024	$\phi(n) \log(\cdot)$, nonparametric	time λn , education xn	deflates c only, OECD	ever born, once
Daruich–Kozlowski 2020	$b(n)u(c_k)$, concave	$C(h, n)$ rising in wage	none	ever born, at fixed age
Couillard 2025	general $U(n, q, c, h)$	goods ni , housing nrh per child	none	number of children
Moreno–Maldonado–Santamaría 2024	multiplicative taste η^i	earnings penalty by timing	none	timing choice
Scholz et al. 2006	none (no fertility choice)	scale only	multiplies and deflates	at home
Borella et al. 2023	none	scale only	deflates	at home
This model	ψm , linear, per period at home	deflating scale; step floor h_P ; one-time ξ	deflates only	n and m

Not on disk and cited from memory only: Becker and Lewis (1973), Greenwood, Seshadri and Vandenbroucke (2005), Adda, Dustmann and Stevens (2017), Guner, Kaygusuz and Ventura (2020), Jones, Schoonbroodt and Tertilt (2010), Kleven et al. (2019). Verify before any of them appears in the paper.